



9/1/2026

N + - Network Fundamentals & Building Networks.

Software-and-Hardware-Tools.

TASK 1.

- ❖ **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

➤ **Wireshark Traffic Capture:**

Procedure:

- Open Wireshark and select the active network interface.
- Start capturing network traffic.
- Browse Instagram or YouTube for about 2 minutes.
- Stop the capture and save it as a .pcap file.

Top 3 Protocols Observed:

- TCP
- TLS
- DNS

Result: The network traffic was captured for 2 minutes and saved successfully as a .pcap file.

TASK 2.

- ❖ Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

➤ Compare Ping Response Times:

Command Used:

```
ping www.spotify.com  
ping www.zomato.com
```

Result:

Website	Response Time
Spotify	21ms
Zomato	14ms

Conclusion: The website with the lower response time responds faster.

TASK 3.

- ❖ **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

➤ **Running Processes and Memory Usage:**

Process Name	Memory Usage
Google Chrome	245 MB
Chrome – Tab	180 MB
Chrome – GPU Process	120 MB
WhatsApp	210 MB

Result: The running Chrome and WhatsApp-related processes were identified, along with their current memory usage.

TASK 4.

- ❖ **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network, Make sure your firewall allows scanning, and only scan your own home/college network.**

➤ **Local Wi-Fi Network Scan:**

I used Advanced IP Scanner to scan my local Wi-Fi network.

IP Address	Device Name
192.168.1.1	Wi-Fi Router
192.168.1.102	My Laptop

Result: Two devices connected to my local network were identified successfully.

TASK 5.

❖ **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

➤ **Hardware Tools for Network Troubleshooting:**

Tool	When I Would Use It
Cable Tester	To check whether an Ethernet cable is working correctly or has a wiring fault.
Crimping Tool	To attach an RJ-45 connector to an Ethernet cable when making or repairing a network cable.
Multimeter	To check voltage, continuity, or power-related issues in network equipment and cables.